CHANDUBHAI INSTITUTE OF SCIENCE AND TECHNOLOGY

Department of Artificial Intelligence & Machine Learning

QUOTATION FOR MVP DEVELOPMENT

Project: EXHYTE Edge Agent for Autonomous Scientific Reasoning in Deep-Space Telemetry (reaction-wheel diagnosis MVP)
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From: Department of Artificial Intelligence & Machine Learning, Chandubhai Institute of Science and Technology
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To: Project Mentor / Review Committee

1. Objective and Scope

This quotation covers a Minimum Viable Prototype (MVP) of an edge-first EXHYTE agent that does not merely detect anomalies in spacecraft telemetry but explains them. The agent implements a closed Explore → Hypothesise → Test loop: a lightweight detector flags an anomaly, an AI (language-model) agent reasons about the mechanism that caused it and generates ranked root-cause hypotheses, each hypothesis is verified against a simplified digital twin, and the agent reports a diagnosed cause to the operator in natural language. Low-confidence or safety-critical cases are escalated to a ground server for human review.

The problem is established and actively researched — reaction-wheel faults are a leading cause of spacecraft attitude-control failure, and machine-learning fault detection is a current (2024–2025) research area. What the existing literature does not do, and what this MVP contributes, is the closed verification loop: generating a mechanistic explanation and testing it in simulation before presenting it to a human, rather than emitting a fault label from a classifier.

The MVP targets a single subsystem — attitude-control reaction wheels — demonstrated on a bench analogue that produces real hardware telemetry. It runs offline (ground-based analysis), with the agent diagnosing and reporting while a human remains in the loop for any action. This prototype targets lab-scale demonstration and is **not intended as flight-qualified hardware**.

2. What the MVP Demonstrates

A reaction wheel is a motor-driven flywheel that points a spacecraft; its bearings wear over years and the degradation appears in telemetry before failure. The bench analogue reproduces this in miniature: a motor spins a flywheel, we induce a fault (increased bearing friction, or a control-gain error), and the agent must not only detect the anomaly but identify which mechanism caused it — the distinction that current detection-only systems do not make.

- **Detection:** a lightweight model flags the anomaly from the live signal.
- **Hypothesis:** the agent generates ranked mechanistic causes (e.g. bearing-friction increase vs. control-gain deviation).
- **Test:** each hypothesis is run against the digital twin; the one reproducing the observed signature is the diagnosis.
- **Report and escalate:** a plain-language diagnosis, with low-confidence or critical cases escalated for human review.

3. Bill of Materials and Cost Estimate

Component prices assume prevailing Indian supplier conditions as of July 2026. Section D is an optional upgrade (see note) that is recommended but not required for the base build.

A. Edge Compute Platform

Item	Qty	Est. Cost (INR)
Raspberry Pi 5 class single-board computer (4–8 GB RAM)	1	15,000 – 23,000
Accessories: PSU, microSD, cooling, case, cables	1 set	3,500 – 5,000

Item	Qty	Est. Cost (INR)
Subtotal A		**18,500 – 28,000**

B. Sensor Front-End

Item	Qty	Est. Cost (INR)
ESP32 WiFi/Bluetooth dev board	2	1,000
MPU6050 6-axis IMU module (vibration channel)	2	500
Wiring & prototyping materials	1 set	500 – 800
Basic mounting / enclosure	1 set	1,000 – 1,200
Subtotal B		**3,000 – 3,500**

C. Prototyping, Cloud & Contingency

Item	Qty	Est. Cost (INR)
Lab accessories & fabrication buffer	–	5,000
Cloud compute / LLM API budget	–	7,000 – 10,000
Subtotal C		**12,000 – 15,000**

D. Reaction-Wheel Analogue — optional upgrade

Item	Qty	Est. Cost (INR)
BLDC motor + ESC (the spinning wheel)	1	2,000 – 3,500
Flywheel mass + shaft mounting	1 set	800 – 1,500
Current sensor (INA219 class) — friction signal	1	300 – 600
Hall/encoder speed feedback — tracking-error signal	1 set	500 – 1,000
Subtotal D		**3,600 – 6,600**

4. Total Estimated Cost

Segment	Estimated Range (INR)
A. Edge compute platform	18,500 – 28,000
B. Sensor front-end	3,000 – 3,500
C. Prototyping, cloud, LLM	12,000 – 15,000
Base MVP total (A + B + C)	**33,500 – 46,500**
D. Reaction-wheel analogue (optional, recommended)	3,600 – 6,600
Total with analogue upgrade	**37,100 – 53,100**

Recommended budget request: INR 45,000 *(Rupees Forty-Five Thousand only)* for the base MVP, with a justified upper band to INR 50,000–55,000 covering the reaction-wheel analogue upgrade and a higher-RAM compute configuration.

> **Why the optional upgrade matters.** An IMU alone measures vibration, which flags that something is wrong but cannot separate a bearing-friction fault from a control-gain error — the exact distinction the agent exists to make. The Section D motor, flywheel, current, and speed channels are what let the demo diagnose a cause rather than only detect an anomaly. Detection alone is well-covered by existing work; the diagnosis is the contribution. We therefore recommend including Section D, but the base MVP is fundable and buildable without it.

5. Inclusions and Exclusions

Inclusions

- Hardware listed above (one edge node, two acquisition nodes; reaction-wheel analogue if Section D is approved).
- Integration of the sensor stream with the edge compute platform.
- The EXHYTE loop: detection, AI-agent mechanistic hypothesis generation, and digital-twin hypothesis testing.
- Escalation logic to a ground server for human-in-the-loop review of low-confidence or critical cases.
- A dashboard visualising the signal, detected anomalies, ranked hypotheses, and the twin-comparison result.

Exclusions

- Flight-qualified or radiation-hardened hardware.
- Integration with actual spacecraft systems or proprietary operator networks.
- Autonomous action on the spacecraft — the agent diagnoses and reports; any corrective action remains with a human operator.
- Additional subsystems or multi-node constellations (scoped separately).
- Long-term production hosting beyond the MVP period.

6. Validity

This estimate is based on prevailing component prices and typical shipping and tax conditions in India as of July 2026 and is valid for 60 days from the date of issue, subject to component availability and market fluctuations.

Prepared by	**Approved by**
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